

Alberto Biscalchin

Applied AI and Data Science

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Professional profile

Research Assistant at Malmö University with an MSc in Computer Science specialising in Applied Data Science and a BSc in Computer Science and Automation Engineering. Experience designing agentic AI systems, analysing human behavioural data and developing reproducible computational pipelines. Combines applied machine learning and statistical evaluation with backend prototyping, decision-support systems and teaching experience.

Research experience

Research Assistant

Malmö University, Department of Technology and Society, Sweden | 2024-present

- Conduct research on AI-supported analysis of travel behaviour, mobility decisions and urban planning data, including large-scale surveys and demographic differences.
- Develop Python pipelines for data cleaning, feature selection, statistical analysis, model-supported reasoning, retrieval and report generation.
- Design modular agentic systems with specialised components for preprocessing, analysis, classification, validation and reporting.
- Use structured interaction traces, shared workflow states, validation checkpoints and intermediate outputs to support interpretability, auditing and reproducibility.
- Contribute to study design, expert- and participant-facing protocols, rating instruments, statistical validation and transparent reporting of limitations.
- Develop retrieval-augmented systems for urban planning and mobility decision support; co-supervise a Master's thesis on multimodal RAG for travel behaviour analysis.

Master's thesis

Leveraging LLMs for Understanding Travel Behaviour

A Case Study on Older Adults in Malmö | Malmö University | 2025

- Designed and implemented a modular multi-agent foundation-model system for large-scale travel behaviour analysis, with structured JSON handovers between specialised agents.
- Evaluated the system in a blinded study with six domain experts against a commercial single-agent system and a human-authored expert baseline.
- Analysed correctness, reasoning, relevance and clarity ratings using one-way ANOVA and Tukey HSD post-hoc testing.
- Received grade A, the Thesis of the Year award and the Wollmarska fonden 2026 scholarship. The work was developed into a peer-reviewed conference paper.

Research interests

Trustworthy and reproducible AI, human–AI collaboration, computational behaviour modelling and visual analytics. Interested in extending current work towards multimodal behavioural sensing, eye tracking and reinforcement learning, including how people adapt their strategies, attention and trust when interacting with autonomous AI systems.

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Education, projects and technical expertise

Education

MSc in Computer Science | Applied Data Science

Malmö University, Sweden | 2025 | 120 credits

Master's thesis: grade A. Relevant coursework: Advanced Machine Learning (A); Big Data Analytics on Cloud Computing Infrastructures (B); Statistical Methods for Data Science (B); Research Methods of Computer Science and Fundamental Computational Theory (B).

Additional second-cycle coursework in Computer Science

Malmö University, Sweden | 2025-2026 | 30 ECTS

IoT Prototyping and User Experience (15 ECTS, A); Introduction to IoT (15 ECTS, B).

BSc in Computer Science and Automation Engineering

Università eCampus, Italy | 2022 | 180 ECTS

Three-year degree, Department of Computing and Control Systems.

Selected projects

Multimodal RAG for travel behaviour analysis

Contributed to a hybrid retrieval system combining layout-aware text processing with region-based visual retrieval. Evaluation used a 487-question benchmark covering correctness, faithfulness, latency, token efficiency and inter-rater agreement.

MAU Advisor

Developed a retrieval-augmented platform for urban planning support, combining structured knowledge retrieval, backend services, domain modelling and user-oriented reporting.

AI-assisted learning pilot study

Designed and facilitated a study comparing conventional reading with AI-supported study using NotebookLM. Used timed activities, open-ended questions and analysis of learner responses and study strategies.

Evidence synthesis

Led PRISMA-guided screening and data extraction for a systematic mapping of LLMs in travel behaviour research. Contributed to a longitudinal review of AI-assisted learning, examining learner regulation, strategy development, dependency and long-term learning trajectories.

Technical and methodological skills

Programming and ML: Python, PyTorch, Transformers, Pandas, C/C++, MATLAB; classification, feature engineering, benchmarking, model evaluation and error analysis.

Agentic AI: Multi-agent architectures, foundation models, prompt-controlled agents, RAG, structured tool use, intermediate-state validation, OpenAI API, LangChain and CrewAI.

Data and backend engineering: SQL, PostgreSQL, FastAPI, Qdrant, CSV/JSON processing, schema-aware pipelines, Docker, Git, GitHub and CI/CD.

Statistics and evaluation: Descriptive statistics, one-way ANOVA, Tukey HSD, inter-rater agreement, controlled comparisons, expert evaluation and reproducible reporting.

Human-centred methods and visualisation: Participant studies, Likert-scale instruments, human-in-the-loop validation and research ethics awareness; QGIS, Matplotlib, spatial analysis, multimodal retrieval and visual reporting.

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Publications, professional experience and references

Peer-reviewed conference papers

Biscalchin, A., Sarkheyli-Hägele, A., Eriksson, J., Vogel, B. (2025). *Designing Education for the AI Era: Principles for Integrating LLMs in Pedagogy*. IEEE SoftCOM, Split, Croatia.
doi: 10.23919/SoftCOM66362.2025.11197458

Biscalchin, A., Sarkheyli, E., Sarkheyli-Hägele, A. (2025). *Multi-Agent Foundation Models for Urban Mobility: The Malmö Elderly Case*. FLLM, Vienna, Austria.
doi: 10.1109/FLLM67465.2025.11391006

Preprint and manuscripts in preparation

Biscalchin, A., Sarkheyli-Hägele, A., Sarkheyli, E., Lord, J., Gołdyka, K. (2026). *Large Language Models in Travel Behaviour Research: A Systematic Mapping of Applications, Data, and Emerging Practices*. Preprint manuscript.

Biscalchin, A., Sarkheyli-Hägele, A., Eriksson, J., Vogel, B., Björklund Boistrup, L. (2026). *The Temporal Paradox of AI-Assisted Learning*. Manuscript in preparation for submission to British Journal of Educational Technology.

Biscalchin, A., Spalazzese, R., et al. (2026). *A Reproducible Pipeline to Discover Serious Games*. Manuscript in preparation for submission to Information Systems.

Teaching and professional experience

Teaching Assistant | Malmö University, Sweden | 2024-2025

Supported undergraduate teaching and student learning activities.

Secondary School Teacher | Istituto Juvarra, Italy | 2022-2023

Taught Mathematics, Physics and Computer Science; more than 500 teaching hours.

Head Teacher | Centro Studi Omnibus, Italy | 2021-2023

Mathematics, Physics and Computer Science.

Full-stack Developer | GIGA srls, Italy | 2018-2019

Head of IT Department | Mondadori Mega Store, Italy | 2014-2015

Awards and academic service

Thesis of the Year, Malmö University (2026); Wollmarska fonden Scholarship (2026). Reviewer for *Mobile Networks and Applications*. Conference presentations and research dissemination in applied AI, mobility analysis and AI-supported education.

Languages

Italian: native | English: C1, IELTS Academic overall band 7.0 | Spanish: B2

References

Arezoo Sarkheyli-Hägele | Associate Professor, Malmö University
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